MS1Vtx_avg_clusE_raw

MC vs data · overview · rank 35 / 98


```
rank 35 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    MS1Vtx_barrel_avg_clusE
    MS1Vtx_endcap_avg_clusE
```

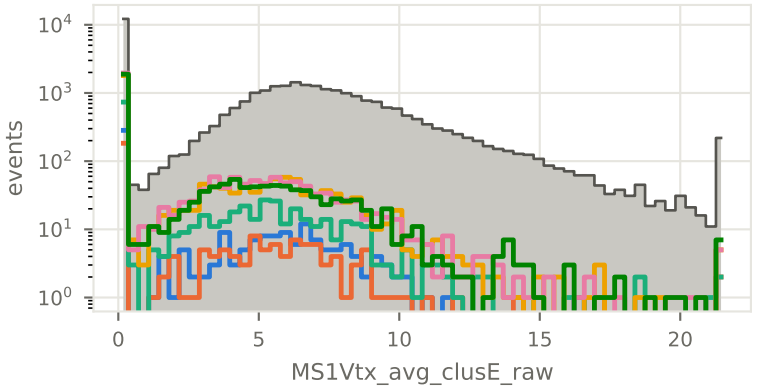
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.361	0.022	0.369	0.357	0.336	0.366	0.366	0.372	0.382	34,970	35/98
barrel	0.304	0.051	0.310	0.333	0.252	0.315	0.323	0.319	0.302	1,089	35/98
all	0.414	0.018	0.421	0.405	0.396	0.419	0.416	0.427	0.429	36,059	17/98

MS1Vtx_avg_clusE_raw

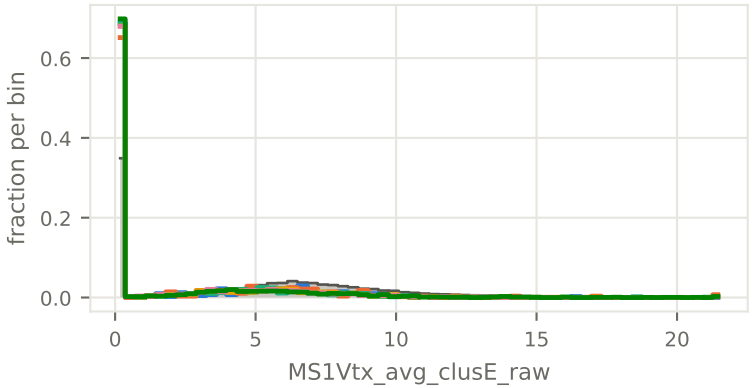
region: ENDCAP · rank 35 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.361 (noise 0.022) D_pooled 0.369 min 0.336 (mH125_mS55) max 0.382 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

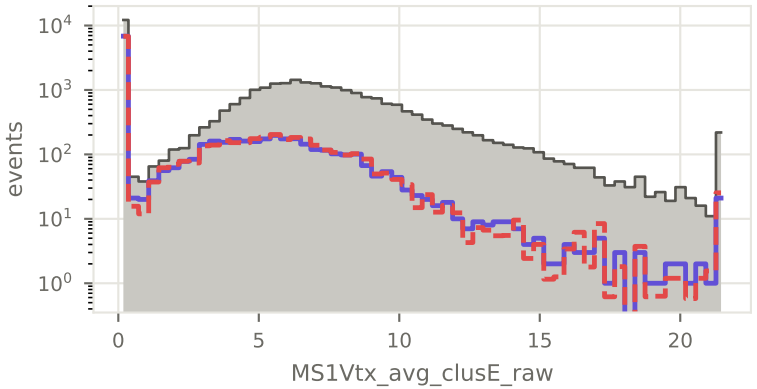


1b · each mass point vs background · normalized

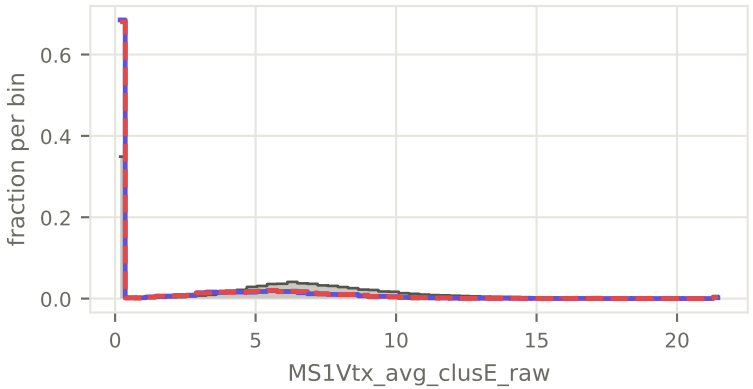


1 · each mass point
background n 34,970
mH125_mS15 D 0.357 n 411
mH125_mS55 D 0.336 n 281
mH200_mS50 D 0.366 n 1,071
mH400_mS100 D 0.366 n 2,648
mH600_mS150 D 0.372 n 2,882
mH1000_mS275 D 0.382 n 2,703

2a · combined signal vs background · raw

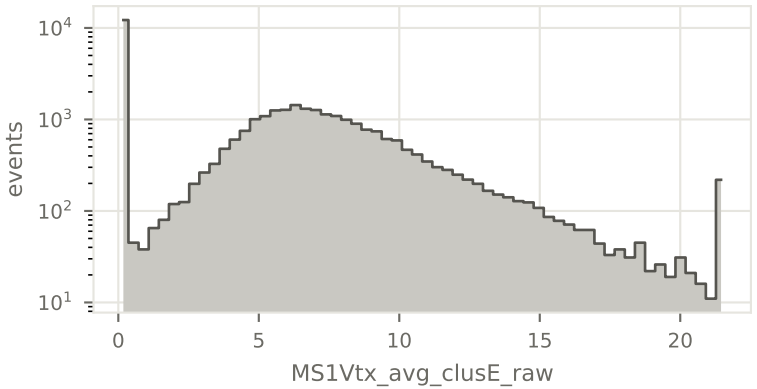


2b · combined signal vs background · normalized

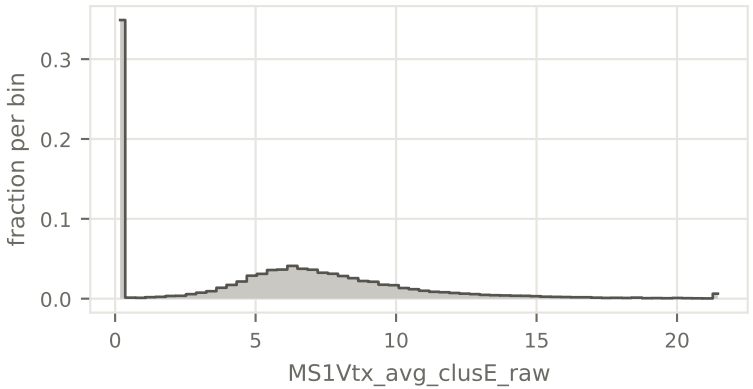


2 · combined signal
background n 34,970
combined, pooled D 0.369 n 9,996
equal weight per mass D 0.361

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

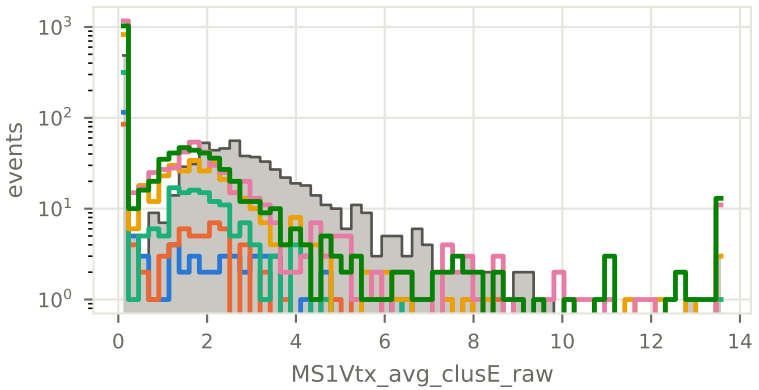
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_avg_clusE_raw

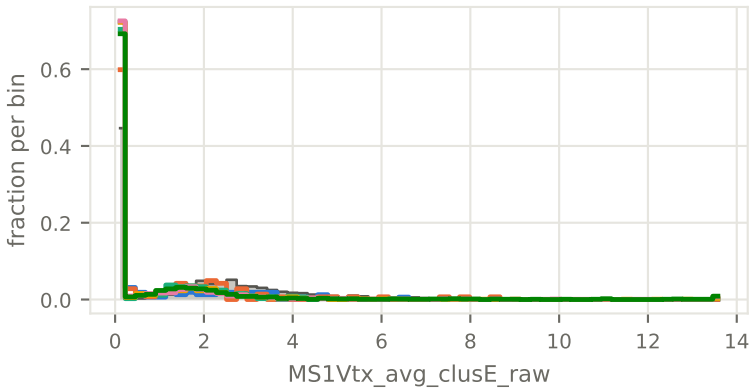
region: BARREL · rank 35 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.304 (noise 0.051) D_pooled 0.310 min 0.252 (mH125_mS55) max 0.333 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

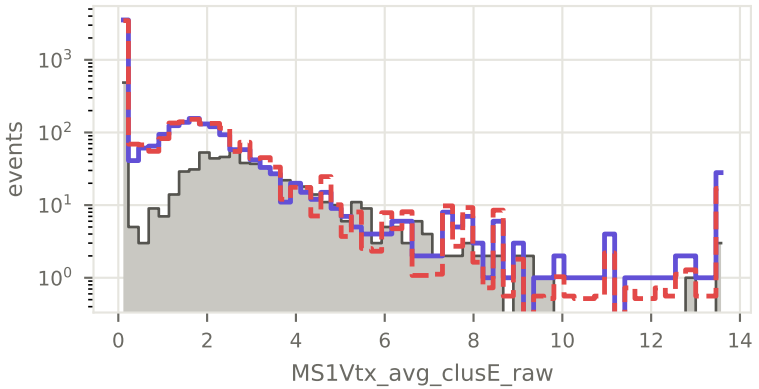


1b · each mass point vs background · normalized

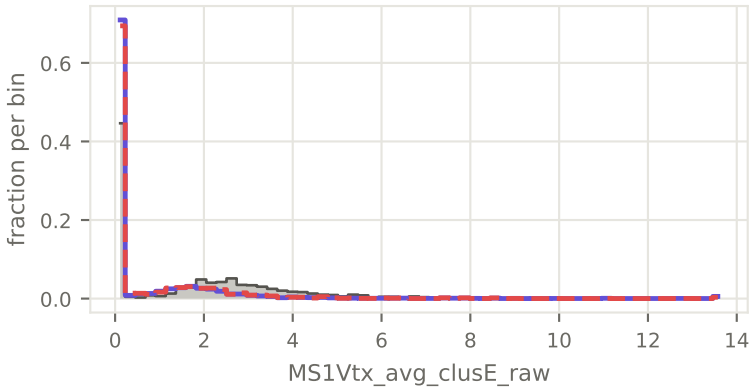


1 · each mass point
background n 1,089
mH125_mS15 D 0.333 n 159
mH125_mS55 D 0.252 n 142
mH200_mS50 D 0.315 n 449
mH400_mS100 D 0.323 n 1,144
mH600_mS150 D 0.319 n 1,608
mH1000_mS275 D 0.302 n 1,485

2a · combined signal vs background · raw

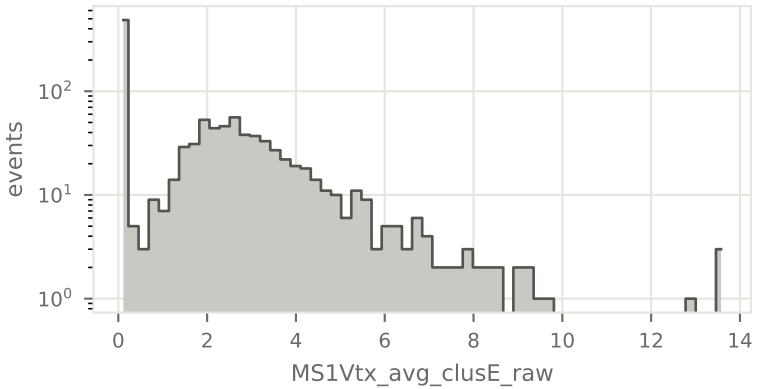


2b · combined signal vs background · normalized

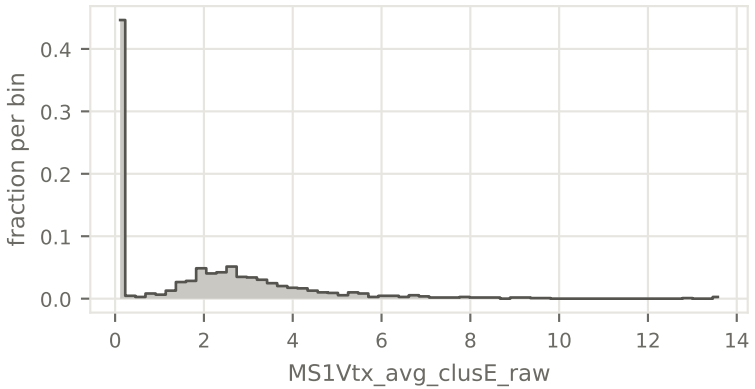


2 · combined signal
background n 1,089
combined, pooled D 0.310 n 4,987
equal weight per mass D 0.304

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

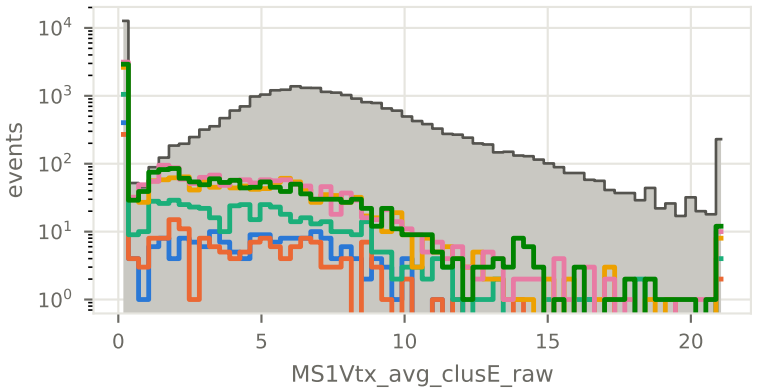
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_avg_clusE_raw

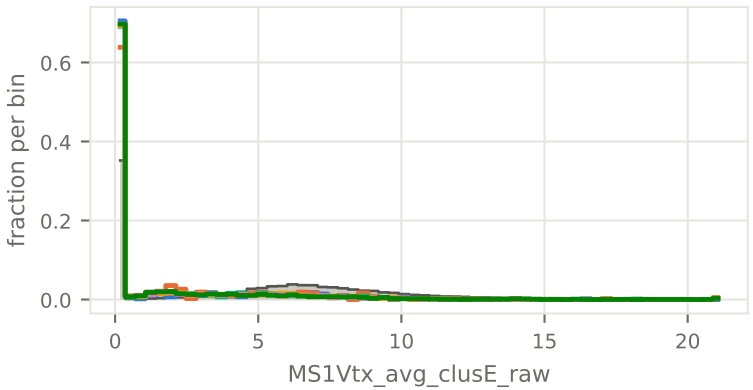
region: ALL · rank 35 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.414 (noise 0.018) D_pooled 0.421 min 0.396 (mH125_mS55) max 0.429 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

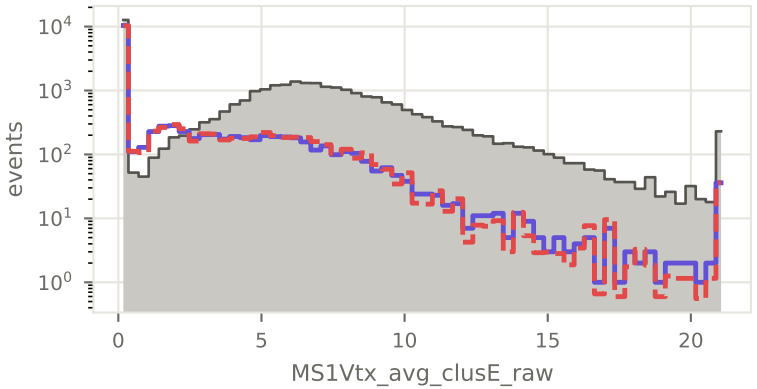


1b · each mass point vs background · normalized

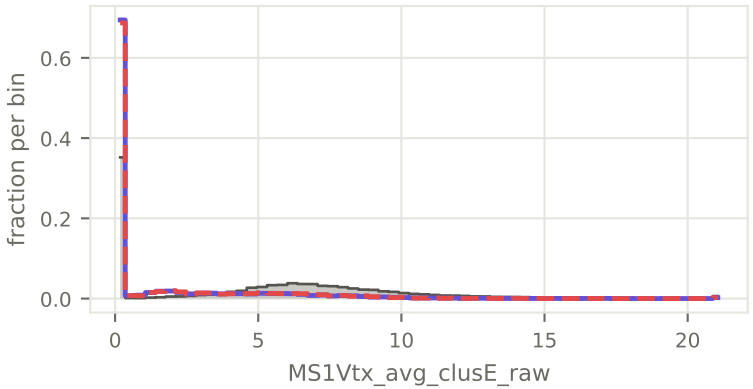


1 · each mass point
background n 36,059
mH125_mS15 D 0.405 n 570
mH125_mS55 D 0.396 n 423
mH200_mS50 D 0.419 n 1,520
mH400_mS100 D 0.416 n 3,792
mH600_mS150 D 0.427 n 4,490
mH1000_mS275 D 0.429 n 4,188

2a · combined signal vs background · raw

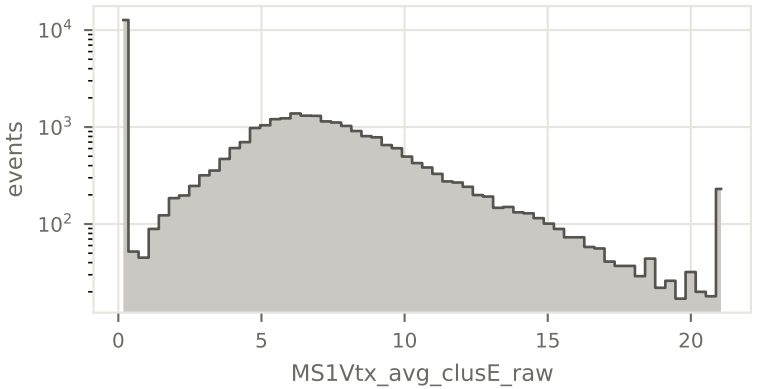


2b · combined signal vs background · normalized

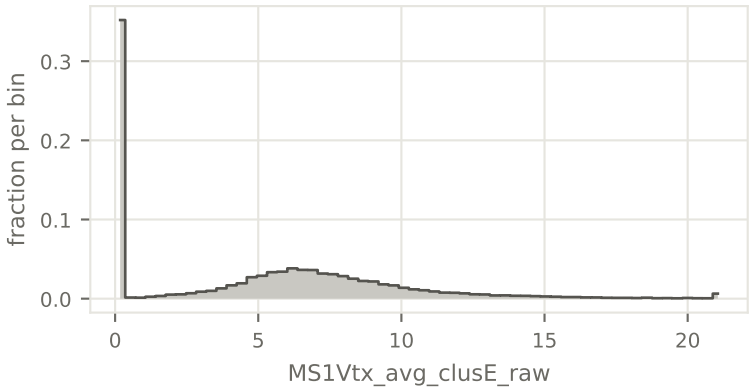


2 · combined signal
background n 36,059
combined, pooled D 0.421 n 14,983
equal weight per mass D 0.414

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed